

EyeRobot 2.0: Active Gaze for Precise Manipulation without Wrist Cameras

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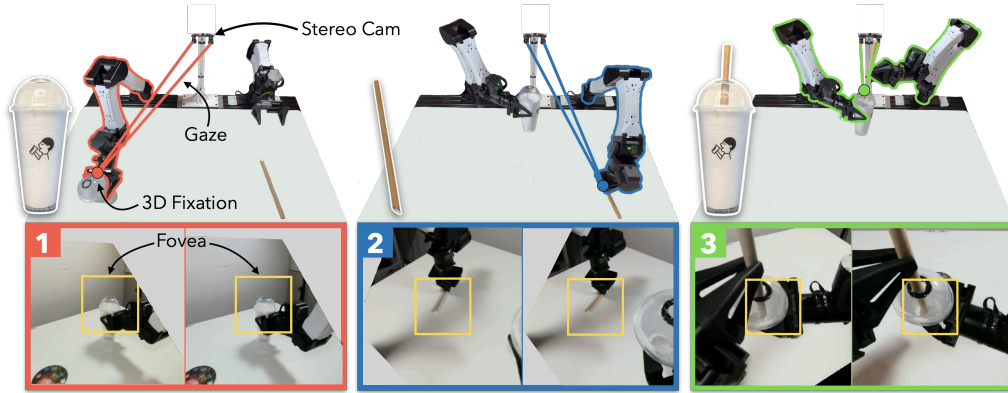


Figure 1: **EyeRobot 2.0** enables fine-grained bimanual manipulation like inserting a straw into a boba cup using only a fixed stereo camera. It coordinates a sequence of gaze targets, for example here first at the boba cup, then the straw, then the insertion point. Gaze targets are output from a semantic object selector, which conditions a low-level gaze policy to servo precisely. Its design draws inspiration from human foveated vision by leveraging multi-resolution image crops to provide high detail where the policy gazes.

Abstract: Inspired by human vision, we introduce a framework using active gaze to enable fine-grained bimanual manipulation with only a single stereo camera. EyeRobot 2.0 physically attends to a 3D *fixation point* in the scene by swiveling two eye viewpoints to center their gaze on it. The resulting images are processed foveally by allocating more visual tokens to the image centers, focusing computation on task-relevant features. Such Active Visual Fixation (AVF) requires carefully coordinated gaze during task execution, which we accomplish hierarchically by first training a low-level gaze servoing policy conditioned on a goal object, then training a target-selector which emits fixation goals based on task progress. Both modules are trained with RL on real-world data: the first is trained with a dense geometric reward and the second co-trains with the BC gripper policy which allows it to discover fixation sequences that can resemble a human’s fixation sequence while performing the task (Fig. 1). AVF further takes advantage of fixation by canonicalizing gripper proprioception and actions into a rotating fixation-relative SE(3) frame, which compacts the size of the action distribution to learn. We collect teleoperation data for 7 real-world and 6 simulated tasks, and conduct over 1000 physical and 1800 simulated robot trials comparing AVF against passive stereo and ego + wrist camera policies trained on the same data. Removing wrist cameras is costly for standard policies: with only passive stereo, real-world success drops from 52% to 27%. AVF closes this gap without any wrist cameras, outperforming passive stereo by 40% in real and 20% in sim. It matches ego + wrist policies when their wrist views are clear (69% vs. 64%), and more than doubles their success when grasped objects occlude the wrist cameras (48%

vs. 22%). These results suggest active gaze as a single-camera alternative to wrist cameras for fine-grained manipulation.

1 Introduction

Consider the task of picking up a cup with a lid, grasping a straw, and inserting the straw into a small opening in the lid (Fig. 1). As humans carry out this task, our eyes flit around the scene, filtering information with coordinated look-ahead movements that anticipate hand motion (Land and Hayhoe [1]). This task-relevant visual sampling, called *fixation*, directs high resolution foveal vision to relevant parts of the visual field. Fixation also provides 3D anchor-points for manipulation, simplifying the challenge of spatially reasoning about the scene [2] and implicitly breaking larger tasks into concrete sub-goals as gaze switches between targets.

We hypothesize that translating this behavior to robot learning could benefit robot manipulation. Fixating on a dynamic task-relevant 3D point acts as a form of physical attention which centers input observations, compacts the data distribution, filters irrelevant visual regions, and allocates additional resolution where it matters for fine-grained tasks without overwhelming with panoramic high resolution. Some of these benefits can be offered by wrist-mounted cameras, which are used by nearly all existing bimanual manipulation systems [3, 4, 5, 6]. However, colocating vision with the gripper prevents decoupling precise visual inputs from the manipulator, which can be critical in tool use, whole body manipulation, or under object occlusions. In addition, these systems constrain manipulator design because of their direct mounting to the gripper, which adds complexity, often occludes the ego camera, introduces motion blur, and limits sleek manipulators. A purely ego-centric behavior cloning approach has potential to ease constraints on end effector design, allow more effective use of stereo-camera data like human manipulation datasets, and improve tasks which benefit from vision-manipulator decoupling like tool use, fast motion, reaching into tight spaces, or searching large spaces [7].

We present a robot learning framework which introduces Active Visual Fixation (AVF) for robotic manipulation. Using input *only* from one fixed stereo camera, AVF can carry out precise manipulation tasks like inserting a straw, capping a marker, zipping a bag, or extracting a wrench from a toolkit, and multi-stage tasks like opening a toaster oven, pulling the tray out, extracting a pan, pushing the tray back, and closing the door again.

AVF couples a fixation policy with a fixation-centric gripper policy. The fixation policy has two modules: a low-level gaze policy that coordinates two eye viewpoints to rapidly find and track 3D objects in the scene, and a high-level target selector to choose *what* object to look at and *when* to look at it during the task. Visual inputs are represented as foveated stereo multi-crops centered at the fixation point allocating more resolution towards the direction of interest. This physical attention not only increases visual acuity, but reduces computation allocated to distractors. In addition, AVF trains a fixation-centric gripper policy via behavior cloning, by representing SE(3) proprioception and action chunks canonicalized into the rotated fixation frame. This effectively constrains the input and output space for the gripper model with respect to task relevant local regions. The gripper policy additionally receives the distance of the 3D fixation point as an observation, which acts as an anchor point for the policy to understand depth. Both the gaze policy and target selector are trained with reinforcement learning (RL), with gaze emerging from a BC-RL loop as in Kerr et al. [7].

We evaluate AVF on 7 physical tasks and 6 simulated MuJoCo tasks, collecting data and training single-task policies from scratch. Across over *1000 physical trials*, AVF outperforms static stereo baselines by 40%, and by 20% on simulated pick-place tasks, with a larger performance improvement on fine-precision tasks. Compared to the widely adopted SOTA ego + wrist setup, AVF *without wrist cameras* matches task performance in physical trials, and on tasks with heavy occlusion of wrist cameras maintains strong performance while wrist baselines regress to static stereo performance. In ablations we find that fixation-centric actions account for a significant portion of AVF’s improvement, causing a 20% performance boost across sim tasks, and foveation contributes strongly to real-world success on fine-grained tasks, boosting by 20% overall.

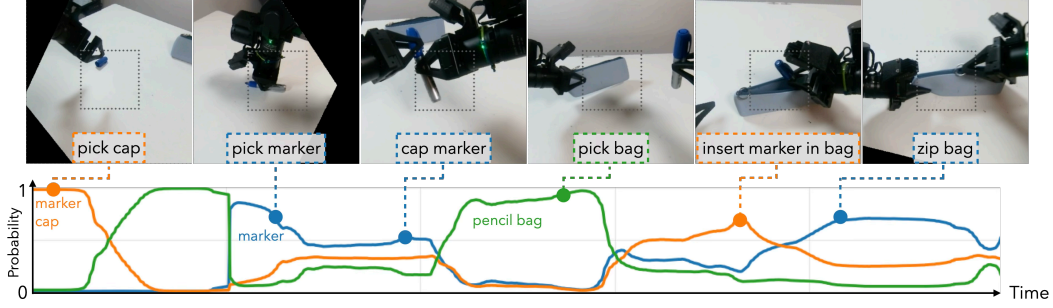


Figure 2: **Gaze target outputs** over an autonomous physical trial which picks up a marker cap and body, caps it, repositions a bag, inserts the marker, and zips the bag. The target selector outputs goals that are relevant to the current stage of the task: for example the policy looks at the marker when capping or the bag when repositioning. Wrist cameras are present but unused.

2 Related Work

2.1 Learning manipulation from human demonstrations

Behavior Cloning (BC) allows robots to learn complex behaviors from teleoperated demonstrations without requiring handcrafted motion primitives and reward functions [8, 9, 10, 11]. BC has been extended to work in complex settings including mobile manipulation [12, 6] and bimanual manipulation [3, 4], and several recent efforts have been directed towards scaling teleoperation datasets [13, 14, 5, 15, 16]. Another enticing direction for scaling is learning from ego-centric human data, either by instrumenting humans or scraping internet videos [17, 18, 19, 20, 21]. However, learning from human videos presents a significant domain gap between ego-centric sensors in first-person datasets and the wrist-cameras that our robot policies so frequently rely on for fine-grained performance. AVF aims to provide a route towards lessening this gap by leveraging human-like fixation for precise bimanual manipulation. We are also inspired by task conditioning approaches in BC which condition a low-level policy on the output of a higher-level task sequencer [22, 23, 24, 25]. Similarly, AVF learns a target selector to control gaze, but does not require human gaze supervision and instead trains gaze with RL via a BC accuracy reward.

2.2 Active vision for manipulation

Active vision deliberately moves sensors to improve observability, filter information, and reduce occlusions for better environment understanding [26, 27, 28, 29, 30, 31, 32, 33]. Previous systems have used active vision to help with tasks such as visual search [34, 35, 36, 37, 38], object detection and tracking [39, 40, 41, 42], and 3D reconstruction [43, 44, 45, 46]. Similarly, active vision has proven beneficial for robot manipulation with a “robot neck” moving the camera based on VR headset movement during task teleoperation to reveal better viewpoints [47, 48, 49, 50, 51, 52]. Active visual search can also benefit from training on ego-centric human data [53, 54, 55]. AVF in contrast doesn’t attempt to address viewpoint selection which has been explored in prior works, but rather investigates how coordinated fine-grained eye movements can boost performance *even with a well-positioned viewpoint*.

Related to learning more precise gaze, some papers collect human eye tracking data during teleoperation to predict a gaze point as an intermediate output to condition a gripper policy with [56, 57] or crop a high-resolution finger-tip image [58]. Others [59] learn gaze in simulation through on-policy task-based rewards, however these methods struggle with sim-to-real transfer. EyeRobot [7] overcomes the need for gaze demonstrations by training a gaze policy on top of real-world behavior cloning data from 360° video by co-training the eye and arm policy with an action feedback loop for reward. Building on this, AVF uses a similar co-training loop but tackles much more fine-grained manipulation tasks by utilizing active stereo fixation, a fixation-centric action representation, and a hierarchical target selector to precisely coordinate gaze throughout execution.

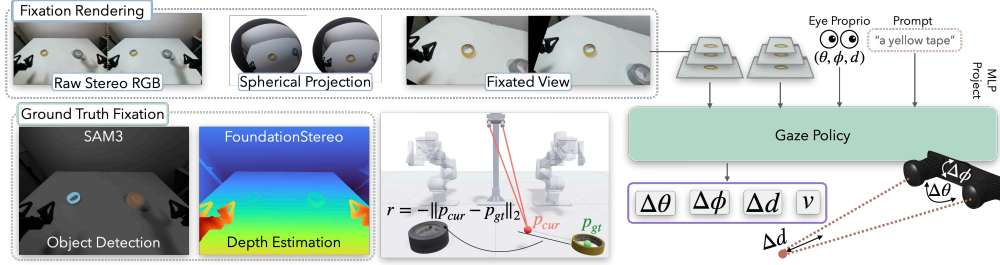


Figure 3: **Learning gaze control.** AVF trains a goal-conditioned gaze policy to continuously servo two eyes’ gaze in 3D. We train this behavior via RL, with a spatial reward derived from detecting the object and estimating its 3D centroid with stereo depth.

3 Manipulation with Active Visual Fixation

Problem Statement: We study visuomotor policy learning from human teleoperation demonstrations for bimanual manipulation with parallel-jaw grippers. Given a set of demonstrations specifying a task, our goal is to train task-specific policies that can reproduce the demonstrated behavior with bounded variations in pose. In particular, EyeRobot 2.0 considers an ego-centric setting in which observations are provided solely through an approximately head-mounted stereo camera. The policy is evaluated using task-specific success criteria.

Overview: AVF trains separate *gripper* and *fixation* policies that work together to achieve a task. The two eyes’ viewpoints actively pivot to converge on a shared 3D fixation point, whose location is updated at each timestep by the fixation policy. The resulting fixation-centered observations are fed into the downstream hand policy, which produces action chunks for robot execution. These SE(3) action chunks are canonicalized into a dynamically shifting fixation frame pointing towards the current fixation point (Fig. 4). Each policy uses a foveated transformer-decoder architecture similar to [7], which we fully describe in the Appendix.

3.1 Learning Gaze Patterns with RL

We train a AVF in two phases. First, we train a goal-conditioned fixation policy capable of fixating on any task-relevant objects. Next, we train a hierarchical target selection policy that sequences fixations to optimize task performance. In the first phase, the fixation policy is trained via a semantic-geometric objective that enables accurate 3D tracking of prompted objects. In the second phase, the target selector is trained with RL in a BC-RL loop as in EyeRobot [7]: a gripper BC policy is co-trained on the resulting fixated views, and the selector is rewarded for fixations that maximize the BC policy’s action prediction accuracy. This hierarchical formulation allows task-dependent gaze behavior to naturally arise without the need for manual human specification.

To learn gaze, we need a way to counterfactually render it on top of existing robot demonstrations that were collected before any gaze policy exists. We therefore need to render what the robot *would have seen* had it looked somewhere. EyeRobot [7] addressed this by collecting demonstrations with a 360° camera and rendering novel views from it. We transfer this idea to stereo, using a calibrated stereo camera to synthesize fixation: rotating a camera about its optical center only requires warping its image, without any 3D reconstruction. This lets AVF use the same stereo camera for data collection, training, and deployment. We simulate fixation by rendering viewpoints from a stereo camera to replicate the appearance of two eyes converging their gaze on a common 3D point in the scene.

Goal-Conditioned Stereo Fixation First, we train a standalone goal-conditioned fixation policy. This policy takes as input a language prompt specifying the goal, the current fixated observations, and the current fixation point in polar coordinates. Though in principle this language prompt could be anything, in practice we limit it to a task-specific annotated set of target objects which the robot interacts with in the scene. The gaze policy outputs local fixation deltas for azimuth and elevation $\Delta\theta$, $\Delta\phi$, and fixation distance Δd represented as a discretized distribution.

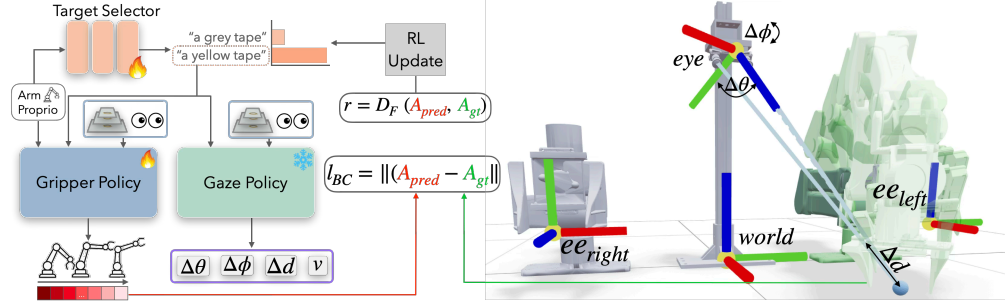


Figure 4: **Sequencing fixation for manipulation.** (Left) AVF trains a lightweight per-task target selector to switch between objects in the scene. This target selector is trained with RL to maximize the prediction accuracy of a gripper BC policy, which is co-trained from scratch at the same time on viewpoints generated from the fixation rollout. Action chunks are supervised with teleop demos, and negative prediction error is supplied as the reward for the selector to optimize. (Right) action chunks are represented as SE(3) chunks in the eye frame, aligned with the current fixation point.

We train this policy with PPO [60] on rollouts from randomly selected stereo still-frames from our teleop demonstrations set. To compute a goal-conditioned reward r for each rollout, we first detect candidate objects by querying SAM3 [61] with the task’s language prompts, then randomly select one detected prompt. To localize the object in 3D, we extract depth from the calibrated stereo pair with FoundationStereo [62] and set the ground truth position to be the 3D centroid of depth intersected with the object mask, p_{gt} . At each step of RL, the policy is conditioned on the current prompt and rewarded based on the current fixation point p_{cur} with $r = -||p_{cur} - p_{gt}||_2$. The result of is a lightweight policy that can rapidly fixate on all task-relevant objects at 30hz.

Learning to Sequence Fixations via a BC-RL Loop While the first phase enables low-level gaze behavior capable of fixating on any object, we now address the high-level decision of **where** and **when** to focus. Like in EyeRobot [7], to circumvent the burden of hand-annotated task stages, we learn this decision process via RL by jointly training gaze alongside the BC policy itself. This naturally encourages gaze sequences that optimize task performance. To do this, we freeze the goal-conditioned fixation policy, and train a lightweight *target selector* with RL, which is conditioned on robot proprioception and outputs a categorical distribution over per-task prompts (Fig. 4, top left). The resulting sampled prompt is used to condition the fixation policy, whose behavior is used during PPO rollout as observations to compute gripper action chunk predictions, which are then used to compute reward for the target selector to optimize (Fig 4, middle). During training, the target selector emits a new target once every 15 frames to allow gaze to settle before sampling a new target.

During train rollouts we uniformly sample a start frame from the demonstration, with special 10% weighting given to the first frame in the demo which often contains the most critical decision for gaze (what to reach for first). The first 30 frames of the agent’s observations operate on a paused frame to allow gaze to settle before being forced to act. We adopt a similar per-step reward as [7], which extracts a task performance proxy from the BC policy’s prediction error. Given predicted and ground truth bimanual action chunks transformed into the same frame A_{pred} , A_{gt} , we first identify an “active arm” whose end effector moves a greater distance in A_{gt} , then compute the arc-length Fréchet distance between A_{gt} and A_{pred} for this arm. The negative of this distance is used as reward to optimize the target selector, encouraging fixation which aids BC prediction accuracy. Limiting reward computation to the active arm avoids sensitivity to spurious correlations in bimanual demonstrations, and Fréchet distance provides a more stable velocity-invariant metric than L2 error.

3.2 Fixation-Centric Action Chunking

Knowing where to focus not only reduces perceptual complexity but also improves action prediction. Specifically, we represent the robot’s proprioception and action chunks relative to the fixation frame. Such a representation standardizes the input about the fixation direction, making the train-

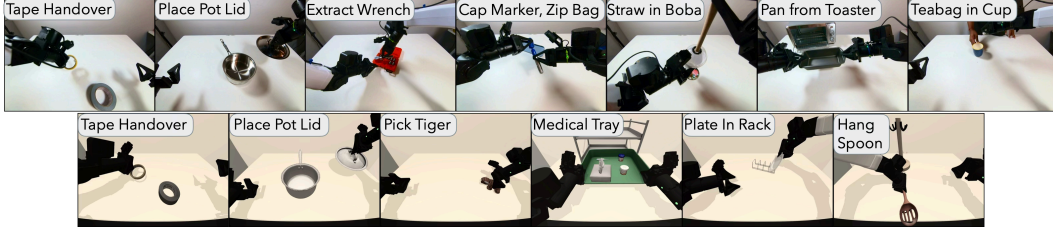


Figure 5: We collect teleop demonstrations for 7 real and 6 simulated tasks. Selected images show the raw perspective of the left stereo camera. Real-world tasks contain precise manipulation requirements, multi-stage behavior, articulated object interaction, and manipulation across the entire workspace. See the Appendix for a full breakdown of task stages and success criteria.

ing distribution more compact. Model input proprioception and actions are represented as SE3 end effector poses transformed into the fixation frame, defined as the cyclopean (virtual center-eye) rotation towards the fixation point in 3D. This canonicalized action representation moves dynamically with fixation, which results in a significant performance improvement over a fixed world-frame. We jointly train the BC policy alongside gaze post-training on perspectives sampled from the fixations induced by RL policy rollout. We maintain a ring buffer of the past 100 rollouts seen, and sample BC train batches independently from this buffer to maintain diversity across demonstrations. The BC architecture is a transformer decoder resembling that of [7], see the Appendix for architecture and hyperparameter details.

4 Experiments

AVF is implemented using a bimanual, 14-DoF I2RT YAM manipulator with parallel jaw grippers. We mount a static stereo camera between the arms 35cm above the base joint, angled 45° downwards for full observability of the table. Each eye captures a 120° diagonal FOV at 1600x1200 resolution. We use a custom leader arm modified from GELLO [63] and TRLC [64] to teleop the robot.

Simulation Environment We build a simulation environment in MuJoCo [65] to conduct repeatable, large-volume experiments. Note that our objective is *not* sim-to-real transfer, nor do we propose to use physics simulation to learn eye gaze, but rather we strive for a digital twin with which to conduct careful method comparisons using the same training pipeline as in real. We replicate the dimensions of the YAM manipulators, and use the public calibrated PD and mass-inertial constants in MJ-Playground [66]. We implement non-pinhole camera rendering in MuJoCo by rendering pinhole images and warping them to exactly match the camera characteristics of each camera. We collect teleoperation with GELLOs through a custom VR interface which streams visuals to the operator in a Quest 3 headset, similar to prior VR teleop systems. All simulation code will be made public.

Tasks and Data In total we collected teleoperation data for 7 physical tasks and 6 simulated tasks as shown in Figure 5. We focus on 5 multi-stage real tasks which involve bimanual coordination (e.g. object handover, coordinated grasping), fine-grained tolerance requirements (e.g. marker capping, zipping, boba straw insertion), and long-horizon behavior (e.g. opening a toaster oven, taking out a pan, pushing the shelf back in, closing the door). Due to the challenges of simulating contact rich physical interactions in real-time, our simulated tasks aim to evaluate a range of basic pick and place behavior. We collect between 10 and 53 minutes of data per-task. Full task definitions, stage visualizations, train distribution visualizations, and dataset size details are provided in the Appendix.

5 Results

We evaluate how well AVF performs compared to SOTA baselines. To construct carefully controlled comparisons, we choose to collect all data ourselves and train all policies from scratch on the *exact same* demonstrations. This avoids obfuscation from unknown pretraining distributions of off-the-shelf models, reduces compute demand, and increases the tractable volume of experiments by an order of magnitude.

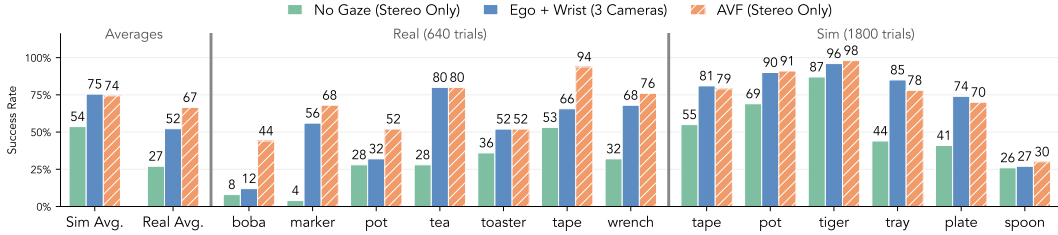


Figure 6: **End-to-end success on real and sim tasks.** Each real task is evaluated on 25 trials for each policy, all sim tasks over 100. Full per-stage conditional success is reported in the Appendix.

In **real**, we train every policy with 3 seeds and randomly select one for evaluation. For each task, we evaluate all policies on 25 scene configurations selected by placing objects within the training data region, but not exactly matching train datapoints. During evaluation, we overlay the live camera stream on the test positions and manually align objects so all policies share the *exact same* test positions. See the Appendix for a visualization of these test distributions. Across all real-world physical trials of baselines and ablations, we evaluate over 1000 policy rollouts.

In **sim**, every baseline and AVF policy is trained across 3 seeds, of which we report the maximum performance to prune failed seeds. See Appendix for full statistics. All sim evaluations are run on the same 100 random configurations, all unseen during training but drawn from the same distribution. Each policy is trained from scratch on single-task data, and executed using the same temporal ensembling weights and inference code. The gaze policy is given 1 second to allow gaze to settle before acting. Per-stage success is programmatically judged using manually specified MuJoCo physics requirements, and each task has a failure timeout based on the average length of demonstrations.

Baselines We collect all our data with a stereo ego and wrist cameras, meaning AVF and baselines share the same training data. *No Gaze* takes in the raw stereo input stream, which has the exact same observability as AVF but without fixation, and *Ego + Wrist* mimics the prevailing bimanual setup using two wrist cameras and the left image from the stereo pair. All baselines are trained with ACT [3] and share the same architecture and training hyperparameters as AVF. We augment baseline input images with 95% center-crop, and 10% translation and rotation randomization to make them less sensitive to placement error, as in [5]. To select the best action space, we test baselines with end-effector (EE) relative SE(3), world-absolute SE(3), joint chunk-relative, and joint chunk-absolute. Since EE-relative SE(3) performs the best in sim (see Appendix), we use that for real evaluations. Baselines are trained with a chunk size of 1 second as suggested in ACT [3] and AVF is trained with 1.5 seconds. We found a slight performance gain of 1.5s for AVF but no such gain for baselines when swept in simulated tasks.

How does AVF compare to gaze-free stereo? This baseline corresponds to training standard policies with static ego-only stereo inputs without wrist cameras. AVF outperforms static stereo policies significantly across all 13 tasks, on average by +20% in sim and +40% in real (Fig. 6). Ego-only policies have a strong tendency to overfit to the training positions, an effect which can particularly be seen in the *tape handover* task, as data was collected in a regular 8×8 grid, or the *tea* task which requires servoing around the entire workspace. In addition, ego policies lack the visual acuity to competently accomplish precise tasks like marker insertion (4%) and straw insertion (8%). Despite having access to the exact same stereo stream, by foveating on relevant objects AVF is able to achieve 44% and 68% end-to-end success on these two tasks. The failure modes of static stereo policies tend to cluster around task stages that require precision (see Sankey diagrams in Appendix), as well as significantly more overfitting behavior in tasks with wide spatial distribution (tape handover).

How does AVF compare against SOTA ego + wrist policies? On precise physical tasks where the wrist camera maintains full visibility (marker, tea, toaster, wrench), AVF performs similarly (Fig. 6). In tasks where grasped objects occlude the wrist cameras (boba,

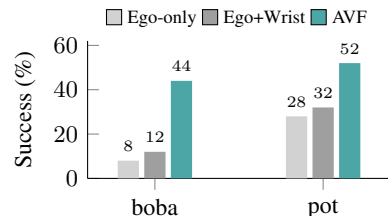


Figure 7: **Wrist occlusion.** When grasped objects block the wrist cameras, ego+wrist falls to near the stereo baseline while AVF maintains performance.

Policy	wrench		tea		tape	
	1st grasp	success	1st grasp	success	1st grasp	success
Ego-only	3/25	0/25	3/25	0/25	6/25	0/25
Ego + Wrist	10/25	4/25	11/25	4/25	14/25	0/25
AVF	20/25	5/25	13/25	4/25	23/25	8/25

Table 1: **Real-world performance with distractors.** EyeRobot 2.0 is better able to ignore distractor objects because of its physical attention directed towards relevant objects.

Policy	boba	marker	pot	tea	toaster	tape	wrench	Mean
AVF –foveation	12/25	5/25	10/25	12/25	9/25	47/64	15/25	46.5%
AVF –stereo	10/25	8/25	10/25	15/25	6/25	51/64	16/25	48.5%
AVF	11/25	17/25	13/25	20/25	13/25	60/64	19/25	66.5%

Table 2: **Real-world ablations** show that foveation and stereo inputs are both important.

pot), ego+wrist performance collapses to near the gaze-free stereo baseline, while AVF maintains strong performance, overall a $2\times$ improvement (Fig. 7). In the `tape_handover` task, wrist camera policies tend to overfit to the grid training distribution similarly to ego-only policies because the wrist camera grasping the yellow tape is occluded until just before it places it when yellow tape is already in an un-recoverable position.

Does fixation improve robustness to visual distractors? We evaluate AVF in real on `tea`, `wrench`, and `tape` with colorful distractor objects randomly scattered on the table and report their end-to-end success along with first stage success in Table 1. All comparisons use the exact same object positions to ensure comparability. The wrist camera often reaches for the largest or most distinct object in its view. Once it begins reaching the policy cannot recover since the wrist camera loses sight of the target. In contrast, EyeRobot 2.0 reaches for the correct object when its gaze is correctly directed at it, and mostly misses grasps when it is looking in the wrong place. This behavior is most reflected in first-grasp success rates, where EyeRobot 2.0 has a significantly higher success rate. Decoupling vision and action could improve robustness in cluttered scenes by literally ignoring irrelevant information.

Gaze analysis See Fig. 2 for a visualization of target selector probabilities over the full course of the marker task. AVF first fixates on the marker cap then switches to the marker before it picks and through the cap motion. Overall, AVF learns intuitive gaze switching, for example looking at an object until just after it has been grasped. Interestingly, it sometimes uses static objects as a sort of anchor during manipulation, for example during spans of the marker task it gazes at the pencil pouch which during data collection was always in a roughly straight-ahead position.

5.1 Ablations

No foveated image architecture We ablate the multi-scale image pyramid and train both gaze and gripper policies on one peripheral view for each eye (Tab. 2). Overall this harms performance by 20%, with the largest gap on precise physical tasks which require detailed image understanding. Interestingly, in simulation this ablation performs the same as AVF. This could be because of the more accurate “replayability” of simulation, where it demands less closed-loop servoing as the real world.

Fixation-centric actions We test an ablation of AVF in simulation which uses SE(3) actions and proprioception in the *world* frame rather than a fixation-relative frame. This ablation performs on par with the gaze-free ego baseline (-21%). Including gaze while still allowing a global output space for the policy seems to still afford overfitting to specific trajectory positions in this low-data regime.

Stereo vs Mono We train an ablation which only has one input image and no input fixation depth. This ablation performs on average 18% worse than AVF across all real-world tasks, indicating that the model benefits from access to binocular information.

6 Limitations and Future Work

AVF currently only trains task-specific gaze and gripper policies. An interesting direction for future extension is to investigate how to adapt the framework to maintain multi-task success performance. The usage of a target selector means that gaze would need to be trained on a sufficiently large corpus to allow generic prompts, which may necessitate different architectures of generating prompts such as a vision-language model. Separately, in this work we do not control head or neck movement, and only proposes how to coordinate fine-grained eye movements provided a good head position is already available. Finally, while object-level pretraining is sufficient for many manipulation tasks, AVF cannot yet train sub-object-level gaze which may provide more benefits for manipulation.

7 Conclusion

This work presents EyeRobot 2.0, a robot learning framework for fine-grained bimanual manipulation using only a fixed stereo camera that leverages active visual fixation to direct attention towards task relevant features. Our experiments suggest that AVF more effectively leverages stereo data than non-active stereo baselines, strongly outperforming it across a range of high-precision physical tasks and matching or exceeding the performance of wrist cameras. Ablating any component of AVF results in a performance drop, indicating it utilizes multi-resolution image inputs, fixation-centric actions, and stereo information to effectively facilitate fine-grained real-world manipulation. This suggests exciting potential for systems which use active visual fixation for fine-grained manipulation instead of wrist cameras.

Appendix

A Additional Results

Failure modes analysis Please see the included Sankey diagrams (Figs. 9, 10, 11) for a visualization of per-stage success breakdown for AVF and baselines across all physical trials. The primary failure cases align with the most difficult, precision-requiring stages of each task, for example inserting the boba straw or capping the marker. In comparison, the ego only baseline struggles more with these precise stages, and also has considerably more failures in “easy” portions of each task, for example picking up objects initially. With only a fixed input view the policy struggles to achieve the precision required to succeed, and even struggles more at basic picking, for example in the tea task only grasping the small teabag 32% of trials, while AVF always grasped the teabag. The ego+wrst baseline recovers this grasping precision, but fails severely on task stages (pot on lid, boba insertion) where wrists become occluded by the grasped object. Even in tasks with unoccluded views from the wrist camera, wrist cameras have similar conditional success rates as AVF, indicating that failures stem from a common modeling or data insufficiency which wrist cameras provide no advantage for.

How well do policy rankings in sim and real correlate? To measure sim-real correlation we collect 2 similar tasks (pot and tape) in sim and in real. While the absolute success rate in sim is not predictive of real-world success, policy rankings are maintained on these tasks between all baselines, and results indicate that sim may favor wrist cameras over physical trials, perhaps because EE-relative chunks are more sensitive to compounding action errors with imperfect real physics than simulated. Pot real-world success is notably lower overall because simulation has more forgiving physics for pot lid closure.

What fixation patterns does AVF learn? Probabilities while inserting the marker into the bag are more ambiguous as all objects are in the same location. Quantitatively, we analyze per-stage gaze in simulation and report full task results in the Appendix. During the first task stage, 100% of policies learned to look at the first object the robot grasps. In `tape_handover`, `dish_rack`, and `tray` the gaze switches to the second object, while in the pot and spoon tasks it remains on the first object.

Does fixation quality correlate with task success? One reasonable null hypothesis could be that the benefit of AVF is simply viewpoint-augmentation over static gaze, for example as shown in Chen et al. [67]. We disentangle these effects by training object-centric-gaze baselines in simulation, one separate policy for each object in the task. Fixating on the *worst object*, as measured by success rate, causes a sharp 24% average decline in performance, and fixating on a single *best* object still results in a 5% drop in performance. Sub-optimal gaze clearly prevents or hampers the policy from completing tasks.

How accurately can AVF fixate in 3D? Overall, the average fixation error after gaze pre-training is 3.5cm computed over 800 random frames from all trajectories. Fixation quality is best seen in our included policy rollout videos, which visualize real-time fixation from the policy during execution. Despite being trained only on still frames, the policy can reliably track moving objects during robot execution.

B Model Architecture and Training Details

Both the fixation and gripper policies use a transformer architecture which decodes actions from input image, proprioception, and goal prompt observations. The input proprioception includes the current fixation point encoded as a unit vector direction and distance. Image tokens are extracted using a frozen DINOv3 [68] ViT-S/16 backbone in foveated manner: a pyra-

	Periphery (cm) ↓	Foveal (cm) ↓
Sim Tasks	4.73	3.51
Real Tasks	4.44	3.55

Table 3: **3D fixation euclidean error** during fixation pretraining, averaged over the final 100 training iterations of the best seed per task.

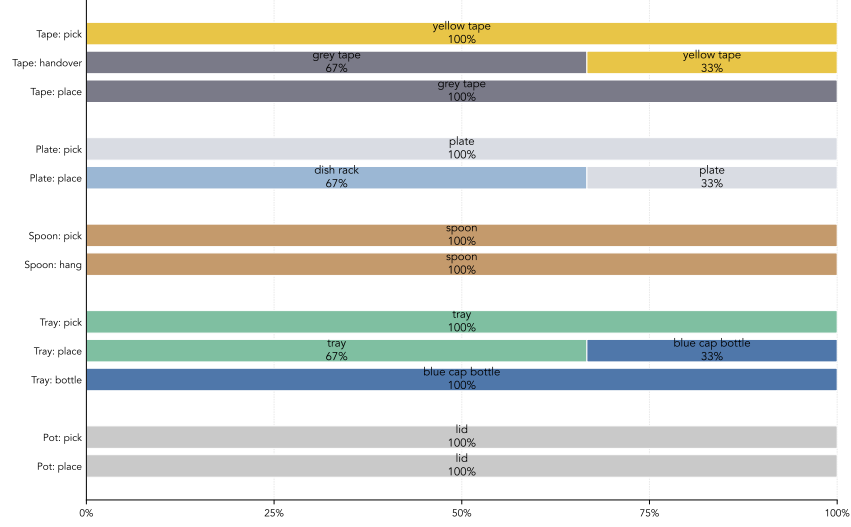


Figure 8: **Per-stage gaze selection in simulated evaluations.** For each policy trained on sim data, we compute the mean gaze pointing error to each object in the scene, and consider the object with the mean lowest error as the “chosen” object during that stage. AVF reliably learns to pre-attend grasping, as evidenced by the fact it looks at the first object 100% of the time. In sim, it sometimes doesn’t look at the second stage of the task i.e the place action, for example in the pot and spoon hanging tasks. The tiger task is omitted as it only has one stage.

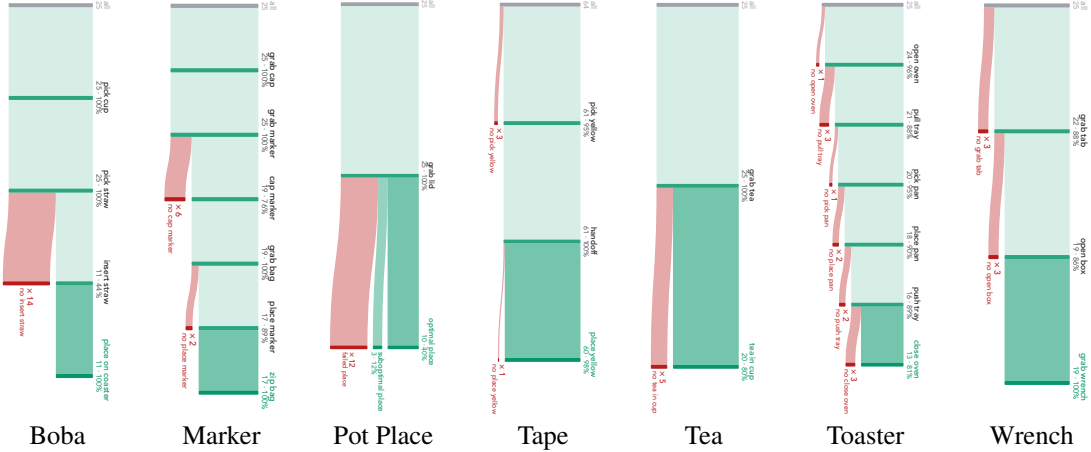


Figure 9: **Per-stage conditional success for AVF** across 25 physical trials per task.

mid of centered crops are independently passed through the encoder and concatenated along the sequence dimension, allowing the model to attend to tokens from multiple resolutions as it desires.

Each input token is given a RoPE [69] (x, y, t) position. For image tokens, x, y corresponds to the location of the token within the global image, and all other tokens are given a value of the image center $(\frac{h}{2}, \frac{w}{2})$. Each token is assigned t corresponding to the current observation timestep, except action chunks which are assigned t consistent with their future time. We use a θ value of 1,000 for sinusoidal frequencies, lower than the default 10,000 for RoPE to adjust for the shorter sequence length. The BC gripper policy takes in history-free single-frame observations, while the gaze policy is allowed to attend to the gaze direction of the previous 3 frames, which stabilizes gaze.

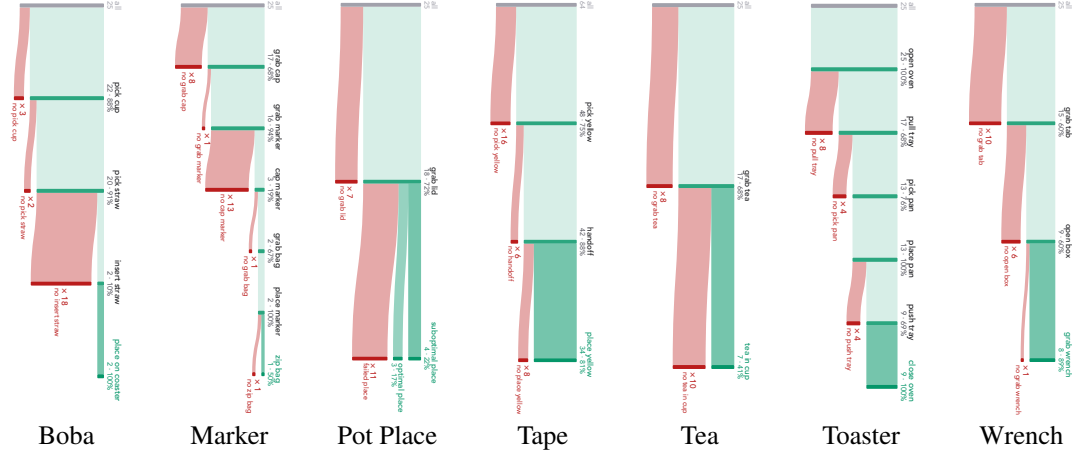


Figure 10: **Per-stage conditional success for Ego Only gaze-free baseline** across 25 physical trials per task.

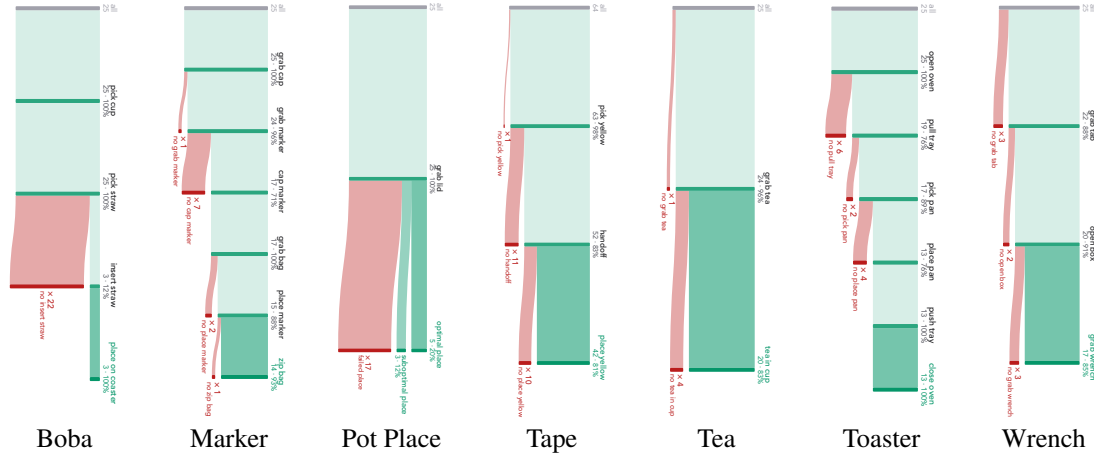


Figure 11: **Per-stage conditional success for Ego+Wrist baseline** across 25 physical trials per task.

The transformer uses bi-directional attention between observation and query tokens, to improve compute efficiency over full self-attention. Each input and output modality is projected to the same token stream and decoded into outputs with a 3-layer MLP. The fixation policy is an actor-critic agent which outputs a categorical distribution over actions alongside a continuous value function estimate v . The target selector is a 3-layer, 256-wide MLP with GeLU [70] activations which takes in proprioceptive input and outputs a categorical distribution over target prompts along with critic estimator. We apply 10% dropout to the hidden layers of the MLP to prevent overfitting to proprioceptive data, along with 10% proprioceptive token drop during BC batch sampling. The attention and FFN hidden layers of the transformer MLPs also have 10% dropout. Each transformer uses 6 transformer blocks, with any learnable query tokens initialized as standard norm embedding vectors with a standard deviation of 0.02 in each dimension. The token dimension of the transformer is 384, and it has an MLP expansion ratio of 4.0, with 12 attention heads. Of each head's 128 dimensions, 32 are used for temporal RoPE, 48 for x , and 48 for y .



Figure 12: **Baseline action space** grid-search over all options in 5 simulated tasks. Note that faded bars are evaluated on the *exact same* positional distributions as the training set to understand train-test overfitting in the simulated environments. Main-paper results are reported with the “Test Set” distribution, which is a different random seed from teleop demonstrations. Across the board, EE-relative (rightmost in each section) outperforms other action spaces, so we use it for all real world baselines. All baseline models exhibit signs of overfitting to the train positions on some tasks (particularly tape and plate), even for tasks where teleop positions were randomly initialized.

B.1 Hyperparameters

All training code will be released for reproducibility, but we report selected important hyperparameters here.

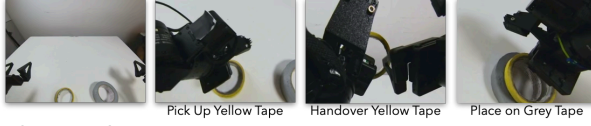
General RL parameters We utilize the standard PPO update rules with value bootstrapping and discount factor $\gamma = 0.95$. We utilize generalized advantage estimation [71] with $\lambda = 0.95$. We do on-policy RL updates for 10 epochs, early-terminating if KL-divergence on the discrete policy output reaches more than 0.02.

The gaze policy discretizes its action space and outputs a cross-product categorical distribution. Azimuth and elevation are binned into 8 polar directions, with 2 magnitudes for each direction (10° and 3°) along with a zero action. Depth deltas are binned with 3 magnitudes (10cm, 5cm, 1cm) and a zero action. The resulting cross-producted action space has 119 bins for the policy output. We prefer discretized action spaces as it helps break ties in ambiguous actions (ie duplicate objects)

Gaze pre-training Each policy rollout consists of 120 timesteps, all over a paused stereo frame. The policy is optimized with the AdamW [72] optimizer, with a learning rate of $1e-4$, $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay of 0.1, with a linear-ramp over the first roughly 5% of train steps and cosine decay over the last 30%. Each rollout uses 64 parallel environments, with a mini-batch size of 16 environment rollouts per minibatch. Policies are optimized for 15M total rollout steps, which corresponds to 1,950 iterations, or 125,000 unique rollouts total. We use an entropy regularization term of 0.01 during training, which decays to 0.005 alongside cosine decay.

Gaze post-training and behavior cloning During BC-RL training, the behavior cloning policy is completely isolated from RL gradients by having its own set of projection and decoder MLPs. Both the target selector and BC policy are optimized with Muon [73], with the learning rate for the BC policy set to $5e-4$ and the target selector $1e-4$. The target selector entropy is regularized with a SAC-style entropy regularizer [74] which tunes the regularization to reach a target average entropy of 70% of the theoretical maximum policy entropy. During post-training, we maintain a buffer of the past 100 episodes of experience, from which we IID sample minibatches of 256 frames for the BC policy to use for SGD. We perform 4 minibatches of BC optimization for each update of the rolling experience buffer. During post-training we roll out 64 parallel environments for 30 steps with paused demonstration time, and 170 steps with time playing forward. During post-training, the video is played at 2x speed (skipping every other frame) so that each episode provides more coverage of the demonstration. We in total rollout 10,000,000 total environment steps, which corresponds to 50,000 unique episode rollouts, 781 iterations, or 3,124 BC gradient steps.

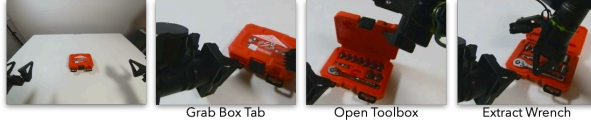
Tape Handover



Place Lid on Pot



Extract Wrench from Toolbox



Cap Marker and Insert in Pouch



Insert Boba Straw in Cup



Take Tray out of Closed Toaster Oven



Place Teabag in Cup



Figure 13: **Real tasks and their stages.** The tasks that we study in AVF represent a suite of complex manipulation behaviors, including **bimanual coordination** (tape handover, grasping and opening a lid), **fine-grained tolerance** requirements (capping a marker, grasping a zipper, inserting a straw into a hole, inserting a marker into a pencil pouch), **challenging visual scenarios** (picking up a white teabag on a white table), and **multi-stage tasks** (6 for both marker and toaster oven tasks).

Training details Each train run uses 8 L40S NVIDIA GPUs, training with data-parallel process coordination with environments evenly split among GPUs. All reported batch sizes are for the total number, not per-GPU. Training is done in `bf16` mixed precision, except RoPE which is computed at `float32` resolution.

Baseline details To decide the action space to use for all baselines, we considered the classically known options of joint-space and end-effector $SE(3)$ space, both chunk-relative and chunk-absolute. While community wisdom has recently drifted towards increasingly using chunk-relative representations, we wanted to ensure that in our data-frugal scenario we thoroughly calibrated the performance of these. Testing such a large grid-search is infeasible in physical trials, so we ran a subset of our simulated tasks with all action spaces (Fig 12). Across the board, we found that end-effector $SE(3)$ relative to the first EE frame in the chunk (“EE-relative”) performed the best, so we use this

Tape Handover



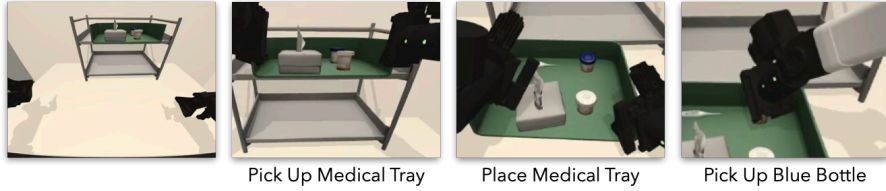
Place Lid on Pot



Pick Up Tiger



Get Medical Tray from Shelf, Pick Blue Bottle



Place Plate in Dish Rack



Hang Spoon on Hook



Figure 14: **Simulated tasks and their stages.** Simulated tasks represent a range of simple pick-place behaviors including single-arm (pot, spoon), single-arm-with-handover (tape), and bimanual coordination (tray, spoon).

action space for all baseline experiments. To doubly verify this, we ran full baseline evaluations with world-relative $SE(3)$ actions on the wrench and toaster tasks, and observed no difference in performance. Baselines are trained at 256 image resolution.

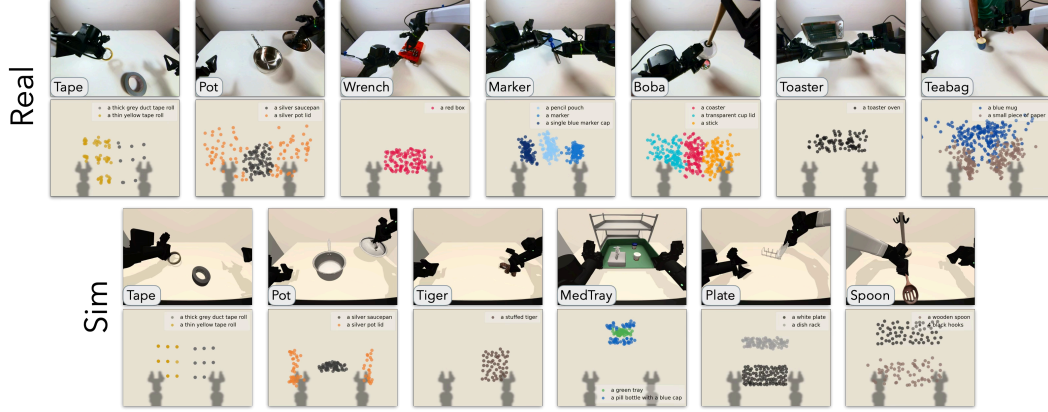


Figure 15: **Training distribution** of objects in the training set. Each dot represents an actual computed ground truth 3D location of an object which has been projected onto the table surface for visualization. Shadows of the YAM arms have been provided for scale. The distribution of objects is approximately uniformly random, except in tape where the distribution is a grid.

C Robot inference

Action chunks are predicted as 30-hz open loop trajectories, which we cubically interpolate at 200hz to obtain the commanded robot positions. We apply temporal ensembling during inference with a k value of 0.01. We additionally compute the temporally-ensembled velocity across predicted action chunks and provide this as a feedforward term for the YAM follower arms, which increases fluidity of the robot movements. See included videos for unedited videos of robot execution.

All models are queried as quickly as compute will allow, with end-to-end throughput of around 40hz for baselines and 20hz for AVF policies.

D Data Details

For stage-by-stage visualizations of all task stages, see Fig 13 and Fig 14. Figure 15 shows the exact training distribution for all sim and real tasks, as calculated by detecting each objects’ starting position in 3D and plotting them projected onto the tabletop surface. As described in the Results section, notice how the tape task is collected in a grid, while other tasks are randomly initialized within a certain region for each object. Baseline policies are surprisingly sensitive to this positional distribution, with a large gap in performance between AVF and wrist policies on this grid-collected task. This gap is lessened by collecting randomly distributed initial states.

See Tab 4 for dataset size statistics for all tasks.

E Simulation Details

We use MuJoCo’s Newton solver with an elliptic friction cone, implicit-fast integration at a 0.5 ms timestep, and 10 noslip iterations to ensure stable contact-rich grasping behavior in simulation. Collision meshes are generated with a convex decomposition from CoACD [75], and we use the default YAM collision meshes and physical properties from MuJoCo Playground [66]. Simulation environments have hand-coded initial object distributions and physical success measures, and the same parameters are used across all models. The same seed is used for environment initializations to ensure direct comparability.

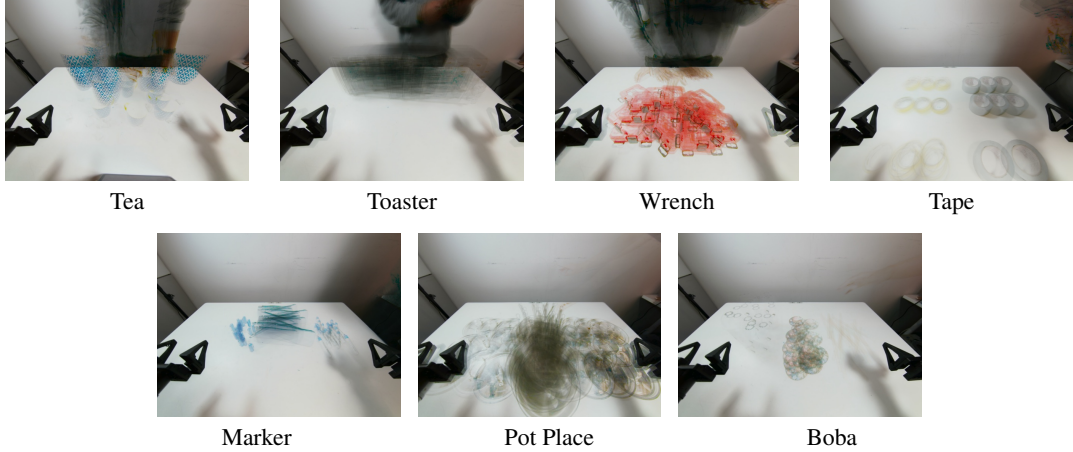


Figure 16: **Test distributions**, visualized by averaging compositing all 25 test overlay images on top of each other. During each evaluation we standardize reset positions across policies using these.



Figure 17: **Subsample of distractor positions** during the wrench task (25 total).

F Evaluation Details

F.1 Test Distribution

See Fig. 16 for a visualization of the test distributions that we use for policy evaluation. All policies are evaluated on the exact same positions, within human placement error, by aligning object positions with an overlay of the current position vs test position.

F.2 Distractor objects

We follow the same procedure for distractor experiments as full evals, by recording 25 positions of test objects and distractor objects and exactly repeating them for all models. See Fig 17 for a random subsample of initial positions of distractor objects in the `wrench` task.

F.3 Per-stage success criteria for real tasks

For all stages, we allow 3-retries before counting a stage a failure, and count a failure if the robot stops moving for more than 10 seconds.

Tape *Pick yellow*: the robot must grasp the yellow and lift it fully off the table without dropping it before the next stage. *Handover*: the robot must successfully transfer the yellow tape to the right gripper without dropping it. *Place*: the robot must place the yellow tape on top of the grey tape without it falling off or tilting. The static position must be stable with the center of gravity fully supported by the grey tape roll.

Pot *Pick lid*: the robot must fully grasp and lift the pot lid off the table without dropping it. *Place lid*: the robot must place the pot lid on the pot **with no gaps visible**. This success criteria is quite strict, as simply resting the lid on top of the pot is insufficient. The lid must fully seal the rim of the pot, which is partially why the real success of this task is significantly lower than the simulated version of the task.

	Simulation						Real							Total
	plate	tape	tiger	hang_spoon	medical_tray	pot_lid	tape	boba	wrench	marker_bag	pot_lid	tea	toaster	
# Demos	100	64	50	64	63	64	64	108	114	126	95	213	89	1278
Minutes	21.6	10.8	4.0	8.1	18.4	7.5	9.5	31.1	26.3	53.4	11.3	26.5	52.0	287.7

Table 4: Per-task demonstration counts and total duration of demonstrations.

AVF (full)	AVF (−fix. relative)	Object gaze (worst obj.)	Object gaze (best obj.)	AVF −stereo	AVF −foveation
0.74	0.53	0.50	0.69	0.74	0.72

Table 5: **Simulated Ablations.** Getting rid of fixation-relative actions (“−fix. relative”) severely harms performance even on relatively simple pick and place tasks, dropping by 21% overall. Restraining gaze to one constant object during training produces a 24% drop for the worst option, and even for the best constant-gaze choice results in a 5% reduction. Interestingly, both stereo and foveation ablations have little effect in simulated tasks, which we hypothesize is because they do not possess the visual complexity or fine-grained tolerance to require precise visual features or spatial understanding.

Toaster *Open door*: the robot must fully open the door to expose the internal oven. *Pull rack*: the robot must pull out the internal wire rack from the toaster oven, and success is only marked if the rack is fully extended to reveal the pan inside, and the tray is not over-extended such to physically prevent pushing it back in. *Pick up tray*: the robot must grab the tray inside the toaster oven with both hands (failure if only one grasps), and gently place it on the table (failure counted if the tray is dropped from more than about 3 inches or twisted during placement). *Push rack in*: the robot must push the wire rack back into the oven, with failure marked if it insufficiently pushes the rack to allow door closure. *Close oven door*: the robot must fully close the oven door and end without touching the door.

Marker *Pick cap*: the robot must pick up the cap and reach the capping position without dropping it (mid-air drop counts as a cap failure). *Pick marker*: the robot must pick up the marker from the table and reach the cap position without dropping it. *Cap marker*: the robot must insert the marker into the cap such that the cap reaches a friction-fit over the marker. If the cap is partially inserted and loosely dangling, this counts as a failure. *Grab bag*: the robot needs to grasp the pencil bag and place it upright close to the robot base (failure if not grasped or if the bag falls over). *Insert marker*: the robot must fully insert the marker into the pencil pouch zipper enclosure, with no part of the marker sticking out. *Zip bag*: the robot must fully zip the bag and let go with both grippers (zipping is considered reached when the robot fully closes the bag, with the zipper within 1cm of the edge of the bag).

Boba *Pick cup*: the robot must lift the boba cup off the workspace without dropping it. *Pick straw*: the robot must pick up the straw without dropping it. *Insert straw*: the robot must insert the straw into the hole in the cup, with success counted if the straw is fully seated in the hole after releasing (partially inside and tilting outside is counted as a failure). *Place on coaster*: the robot must place the cup on the coaster, with the cup resting flat fully supported by the coaster at the end.

Teabag *Pick teabag*: the robot must grasp and lift the teabag off the table without dropping it. *Place teabag in cup*: after releasing, the teabag must be fully contained inside the cup, if it hits the teacup wall and bounces out this is a failure.

Wrench *Grasp box tab*: the robot must grasp the black box tab with the left hand securely without slippage during the lid movement. *Open toolbox*: the toolbox lid must be fully open more than 90°. *Extract wrench*: the robot must fully grasp and lift the wrench out of the toolbox, and reach a steady state above the wrench without dropping the wrench. If the wrench falls before reaching steady state, this is considered a failure.

Policy	Subtask 1	Subtask 2	Subtask 3	Success
wrench	<i>grab tab</i>	<i>open box</i>	<i>grab wrench</i>	<i>complete</i>
Exo (Stereo)	3/25	0/25	0/25	0/25
Ego + Wrist	10/25	6/25	4/25	4/25
AVF	20/25	7/25	5/25	5/25
tea	<i>grab tea</i>	<i>tea in cup</i>	–	<i>complete</i>
Exo (Stereo)	3/25	0/25	–	0/25
Ego + Wrist	11/25	4/25	–	4/25
AVF	13/25	4/25	–	4/25
tape	<i>pick yellow</i>	<i>handoff</i>	<i>place yellow</i>	<i>complete</i>
Exo (Stereo)	6/25	1/25	0/25	0/25
Ego + Wrist	14/25	0/25	0/25	0/25
AVF	23/25	12/25	8/25	8/25

Table 6: **Task success with distractors.** Each cell reports the number of trials out of 25 reaching the corresponding subtask. Success denotes end-to-end task completion. Bold indicates the best policy for each task-subtask.

Policy	Plate			Tape			Tiger		
	med	mn±sd	max	med	mn±sd	max	med	mn±sd	max
BC ego-mono	54	52±5	55	38	38±5	41	76	78±5	83
BC ego-stereo	41	41±1	41	52	52±4	55	86	86±2	87
BC ego+wrist	71	71±3	74	79	79±3	81	96	95±2	96
AVF	67	67±3	70	75	76±2	79	97	97±1	98

Policy	Spoon			Tray			Pot		
	med	mn±sd	max	med	mn±sd	max	med	mn±sd	max
BC ego-mono	22	21±5	26	50	48±3	50	69	69±1	70
BC ego-stereo	23	24±2	26	40	39±6	44	62	64±5	69
BC ego+wrist	27	27±1	27	85	83±3	85	89	89±1	90
AVF	26	27±2	30	77	76±2	78	90	89±3	91

Table 7: **Full sim success statistics (%).** Median, mean, std, are reported over 3 seeds for each policy/task combination. Each evaluation of a seed is done over 100 trials.

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